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## Practical 1

**Aim : Design a Web Application for an Organization with Registration forms and advanced controls**

### WebForm1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeBehind="WebForm1.aspx.cs" Inherits="practical_dummy.WebForm1" %>
<!DOCTYPE html>
<html>
<head runat="server">
  <title>Organization Registration</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f4f6f9;
    }

    h2 {
      text-align: center;
      color: #333;
    }

    form {
      width: 600px;
      margin: auto;
      background: #fff;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 0 10px rgba(0,0,0,0.1);
    }

    label {
      font-weight: bold;
    }

    input[type="text"], input[type="password"], select {
      width: 100%;
      padding: 8px;
      margin-top: 5px;
      margin-bottom: 15px;
      border-radius: 5px;
      border: 1px solid #ccc;
    }

    .btn {
      background-color: #007bff;
      color: white;
      padding: 10px;
      border: none;
      border-radius: 5px;
      width: 100%;
```

```

        cursor: pointer;
        font-size: 16px;
    }

    .btn:hover {
        background-color: #0056b3;
    }

    .result-panel {
        margin-top: 20px;
        padding: 15px;
        background: #eaf7ea;
        border-radius: 8px;
        border: 1px solid #b6e0b6;
    }

    .gridview {
        width: 100%;
        border-collapse: collapse;
    }

    .gridview th {
        background: #007bff;
        color: white;
        padding: 8px;
    }

    .gridview td {
        padding: 8px;
        border: 1px solid #ddd;
        text-align: center;
    }

    .error {
        color: red;
        font-size: 12px;
    }
</style>
</head>
<body>
<form id="form1" runat="server">

<h2>Employee Registration Form</h2>

<asp:ValidationSummary ID="ValidationSummary1" runat="server"
ForeColor="Red"/>

Name:
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>
<asp:RequiredFieldValidator ControlToValidate="txtName" runat="server"
ErrorMessage="Name required" ForeColor="Red"/>

```

<br /><br />

Email:

```
<asp:TextBox ID="txtEmail" runat="server"></asp:TextBox>
<asp:RegularExpressionValidator ControlToValidate="txtEmail" runat="server"
ErrorMessage="Invalid Email" ForeColor="Red"
ValidationExpression="\w+@\w+\.\w+" />
```

<br /><br />

Password:

```
<asp:TextBox ID="txtPassword" runat="server"
TextMode="Password"></asp:TextBox>
```

Confirm Password:

```
<asp:TextBox ID="txtConfirm" runat="server"
TextMode="Password"></asp:TextBox>
```

```
<asp:CompareValidator ControlToCompare="txtPassword"
ControlToValidate="txtConfirm"
runat="server" ErrorMessage="Passwords do not match" ForeColor="Red"/>
```

<br /><br />

Age:

```
<asp:TextBox ID="txtAge" runat="server"></asp:TextBox>
<asp:RangeValidator ControlToValidate="txtAge" MinimumValue="18"
MaximumValue="60"
Type="Integer" runat="server" ErrorMessage="Age must be 18-60"
ForeColor="Red"/>
```

<br /><br />

Gender:

```
<asp:RadioButtonList ID="rblGender" runat="server">
  <asp:ListItem>Male</asp:ListItem>
  <asp:ListItem>Female</asp:ListItem>
</asp:RadioButtonList>
```

<br />

Skills:

```
<asp:CheckBoxList ID="cblSkills" runat="server">
  <asp:ListItem>C#</asp:ListItem>
  <asp:ListItem>Java</asp:ListItem>
  <asp:ListItem>Python</asp:ListItem>
</asp:CheckBoxList>
```

<br />

Department:

```
<asp:DropDownList ID="ddlDept" runat="server">
```



```
<asp:ListItem>Select</asp:ListItem>
<asp:ListItem>HR</asp:ListItem>
<asp:ListItem>IT</asp:ListItem>
<asp:ListItem>Finance</asp:ListItem>
</asp:DropDownList>
```

```
<br /><br />
```

Country:

```
<asp:ListBox ID="lstCountry" runat="server">
  <asp:ListItem>India</asp:ListItem>
  <asp:ListItem>USA</asp:ListItem>
  <asp:ListItem>UK</asp:ListItem>
</asp:ListBox>
```

```
<br /><br />
```

Date of Birth:

```
<asp:Calendar ID="Calendar1" runat="server"></asp:Calendar>
```

```
<br /><br />
```

Upload Resume:

```
<asp:FileUpload ID="FileUpload1" runat="server" />
```

```
<br /><br />
```

```
<asp:Button ID="btnSubmit" runat="server" Text="Register"
OnClick="btnSubmit_Click" />
```

```
<hr />
```

```
<!-- RESULT PANEL -->
```

```
<asp:Panel ID="pnlResult" runat="server" Visible="false">
  <h3>Registration Result</h3>
```

```
  <asp:GridView ID="GridView1" runat="server"></asp:GridView>
</asp:Panel>
```

```
</form>
```

```
</body>
```

```
</html>
```

### **WebForm1.aspx.cs**

```
using System;
using System.Collections.Generic;
using System.Data;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
```

```

namespace practical_dummy
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void btnSubmit_Click(object sender, EventArgs e)
        {
            if (Page.IsValid)
            {
                string skills = "";

                foreach (System.Web.UI.WebControls.ListItem item in cbISkills.Items)
                {
                    if (item.Selected)
                    {
                        skills += item.Text + ", ";
                    }
                }

                DataTable dt = new DataTable();

                dt.Columns.Add("Name");
                dt.Columns.Add("Email");
                dt.Columns.Add("Gender");
                dt.Columns.Add("Skills");
                dt.Columns.Add("Department");
                dt.Columns.Add("Country");
                dt.Columns.Add("DOB");

                dt.Rows.Add(
                    txtName.Text,
                    txtEmail.Text,
                    rblGender.SelectedValue,
                    skills,
                    ddlDept.SelectedValue,
                    lstCountry.SelectedValue,
                    Calendar1.SelectedDate.ToShortDateString()
                );

                GridView1.DataSource = dt;
                GridView1.DataBind();

                pnlResult.Visible = true;
            }
        }
    }
}

```

## OUTPUT

Organization Registration x + Ask Gemini

localhost:44360/WebForm1

### Employee Registration Form

Name: Atharva

Email: atharva@gmail.com

Password: .....

Confirm Password: .....

Passwords do not match

Age: 21

Gender: ☒ Male ☐ Female

33°C Sunny 11:34 21-04-2026

Organization Registration x + Ask Gemini

localhost:44360/WebForm1

Skills: ☒ C# ☒ Java ☒ Python

Department: IT

Country: India USA UK

Date of Birth: April 2026

| Sun | Mon | Tue | Wed | Thu | Fri | Sat |
|-----|-----|-----|-----|-----|-----|-----|
| 29  | 30  | 31  | 1   | 2   | 3   | 4   |
| 5   | 6   | 7   | 8   | 9   | 10  | 11  |
| 12  | 13  | 14  | 15  | 16  | 17  | 18  |
| 19  | 20  | 21  | 22  | 23  | 24  | 25  |
| 26  | 27  | 28  | 29  | 30  | 1   | 2   |
| 3   | 4   | 5   | 6   | 7   | 8   | 9   |

Upload Resume: Choose file AI-Security-....cleaned.pptx

Register

33°C Sunny 11:35 21-04-2026

Organization Registration

Ask Gemini

localhost:44360/WebForm1

Department:

IT

Country:

India  
USA  
UK

Date of Birth:

< April 2026 >

| Sun | Mon | Tue | Wed | Thu | Fri | Sat |
|-----|-----|-----|-----|-----|-----|-----|
| 29  | 30  | 31  | 1   | 2   | 3   | 4   |
| 5   | 6   | 7   | 8   | 9   | 10  | 11  |
| 12  | 13  | 14  | 15  | 16  | 17  | 18  |
| 19  | 20  | 21  | 22  | 23  | 24  | 25  |
| 26  | 27  | 28  | 29  | 30  | 1   | 2   |
| 3   | 4   | 5   | 6   | 7   | 8   | 9   |

Upload Resume: Choose file No file chosen

Register

Registration Result

| Name    | Email             | Gender | Skills            | Department | Country | DOB        |
|---------|-------------------|--------|-------------------|------------|---------|------------|
| Atharva | atharva@gmail.com | Male   | C#, Java, Python, | IT         | India   | 01-01-0001 |

23°C Sunny

Search

ENG IN

11:36 21-04-2026



## Practical 2

**Aim : Create a website using the master page concept.**

### Site1.master

```
<%@ Master Language="C#" AutoEventWireup="true"  
CodeBehind="Site1.master.cs" Inherits="master_text.Site1" %>
```

```
<!DOCTYPE html>  
<html>  
<head runat="server">  
    <title>My Website</title>  
  
    <asp:ContentPlaceHolder ID="head"  
runat="server"></asp:ContentPlaceHolder>  
  
    <style>  
        body {  
            font-family: Arial;  
            margin: 0;  
        }  
        .header {  
            background-color: #333;  
            color: white;  
            padding: 15px;  
            text-align: center;  
        }  
        .menu {  
            background-color: #555;  
            padding: 10px;  
            text-align: center;  
        }  
        .menu a {  
            color: white;  
            margin: 10px;  
            text-decoration: none;  
            font-weight: bold;  
        }  
        .menu a:hover {  
            color: yellow;  
        }  
        .content {  
            padding: 20px;  
        }  
        .footer {  
            background-color: #333;  
            color: white;  
            text-align: center;  
            padding: 10px;  
            position: fixed;  
            bottom: 0;  
            width: 100%;  
        }  
    </style>
```

```

</style>
</head>

<body>
  <form id="form1" runat="server">

    <div class="header">
      <h1>My Master Page Demo</h1>
    </div>

    <div class="menu">
      <a href="Home.aspx">Home</a>
      <a href="About.aspx">About</a>
    </div>

    <div class="content">
      <asp:ContentPlaceHolder ID="MainContent"
runat="server"></asp:ContentPlaceHolder>
    </div>

    <div class="footer">
      <p>© 2026 My Website</p>
    </div>

  </form>
</body>
</html>

```

### **Home.aspx**

```

<%@ Page Language="C#" MasterPageFile="~/Site1.master"
AutoEventWireup="true" CodeBehind="Home.aspx.cs"
Inherits="master_text.Home" %>

<asp:Content ID="Content1" ContentPlaceHolderID="MainContent"
runat="server">
  <h2>Welcome to Home Page</h2>
  <p>This is the home page content.</p>
</asp:Content>

```

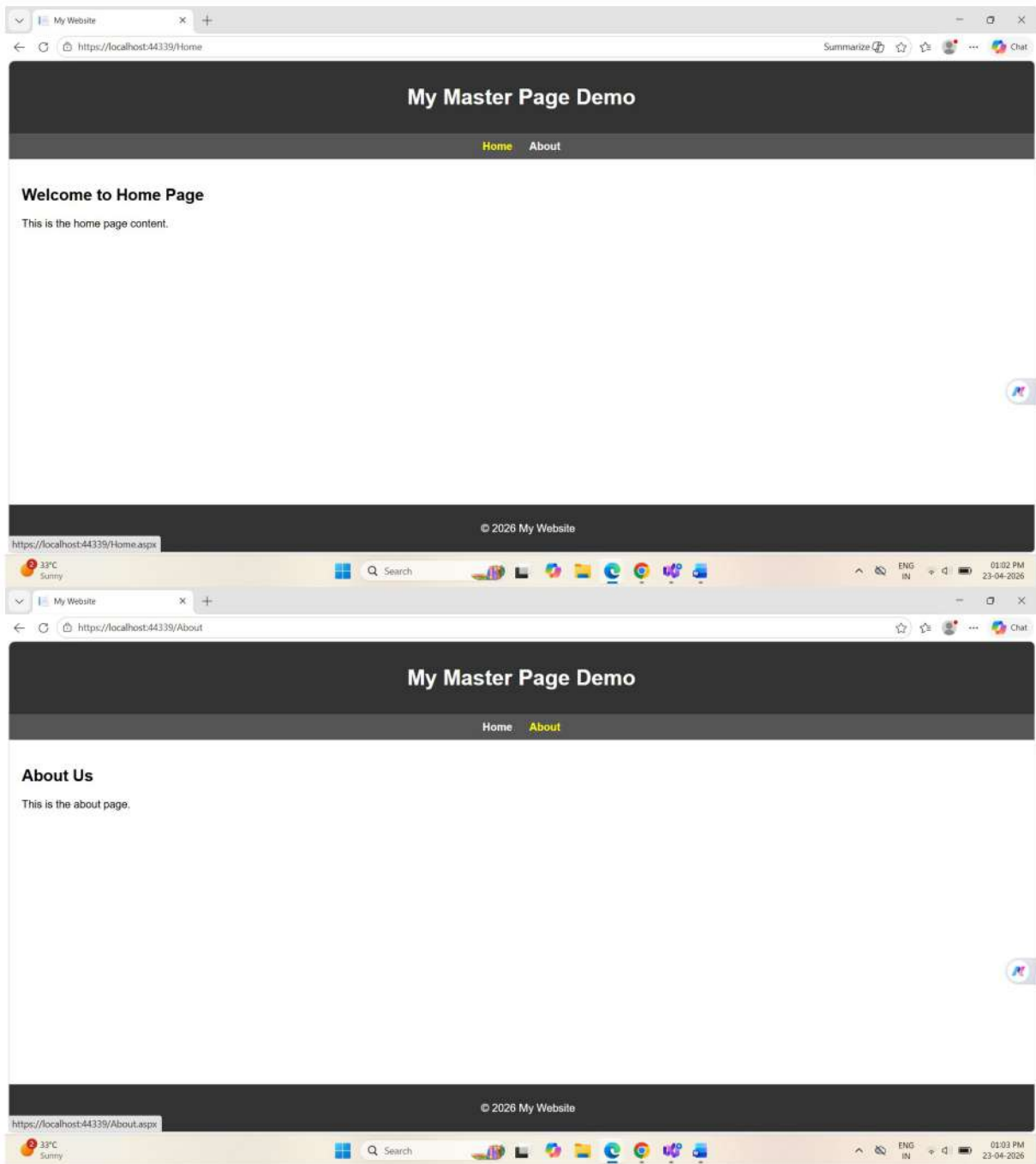
### **About.aspx**

```

<%@ Page Language="C#" MasterPageFile="~/Site1.master"
AutoEventWireup="true" CodeBehind="About.aspx.cs"
Inherits="master_text.About1" %>

<asp:Content ID="Content1" ContentPlaceHolderID="MainContent"
runat="server">
  <h2>About Us</h2>
  <p>This is the about page.</p>
</asp:Content>

```



### Practical 3

**Aim : Design a webpage to demonstrate a connection oriented architecture.**

Webform1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeBehind="WebForm1.aspx.cs" Inherits="boomrang.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:TextBox ID="txtName" runat="server" Placeholder="Enter
Name"></asp:TextBox>
            <asp:TextBox ID="txtEmail" runat="server" Placeholder="Enter
Email"></asp:TextBox>

            <asp:Button ID="btnSave" runat="server" Text="Save"
OnClick="btnSave_Click" />

            <br /><br />

            <asp:GridView ID="GridView1" runat="server"></asp:GridView>
        </div>
    </form>
</body>
</html>
```

#### Webform.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data;
using System.Data.SqlClient;
using System.Configuration;

namespace boomrang
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        string connStr =
ConfigurationManager.ConnectionStrings["DBConnection"].ConnectionString;
```

```

protected void Page_Load(object sender, EventArgs e)
{
    if (!IsPostBack)
    {
        LoadData();
    }
}

protected void btnSave_Click(object sender, EventArgs e)
{
    using (SqlConnection conn = new SqlConnection(connStr))
    {
        conn.Open(); // 🔑 Connection explicitly opened

        SqlCommand cmd = new SqlCommand(
            "INSERT INTO Users(Name, Email) VALUES(@Name, @Email)",
conn);

        cmd.Parameters.AddWithValue("@Name", txtName.Text);
        cmd.Parameters.AddWithValue("@Email", txtEmail.Text);

        cmd.ExecuteNonQuery();

        conn.Close(); // 🔑 Connection explicitly closed
    }

    LoadData();
}

private void LoadData()
{
    using (SqlConnection conn = new SqlConnection(connStr))
    {
        conn.Open();

        SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM Users",
conn);

        DataTable dt = new DataTable();
        da.Fill(dt);

        GridView1.DataSource = dt;
        GridView1.DataBind();

        conn.Close();
    }
}
}
}

```

### Sql commands

```
CREATE DATABASE DemoDB_Connection_01;
```



```
USE DemoDB_Connection_01;

CREATE TABLE Users (
    Id INT PRIMARY KEY IDENTITY,
    Name NVARCHAR(100),
    Email NVARCHAR(100)
);
```

### Web.config

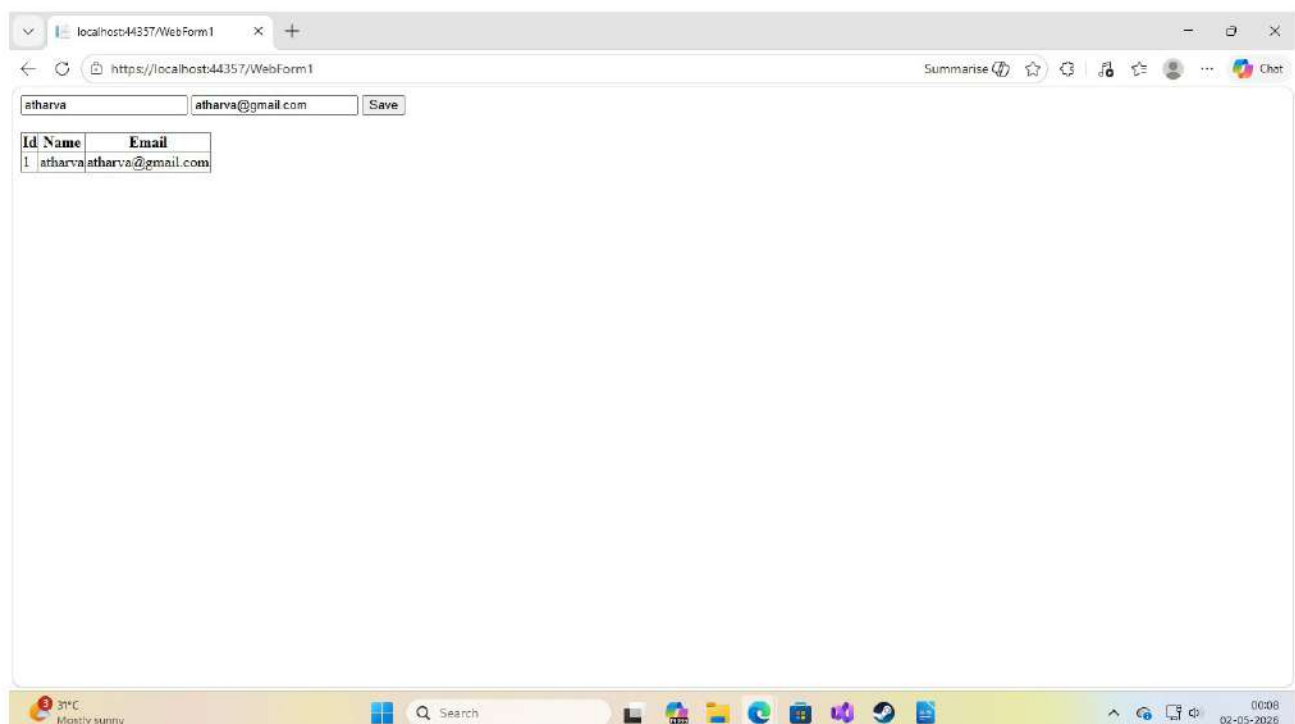
```
<?xml version="1.0" encoding="utf-8"?>
<!--
    For more information on how to configure your ASP.NET application, please
    visit
    https://go.microsoft.com/fwlink/?LinkId=169433
-->
<configuration>
    <connectionStrings>
        <add name="DBConnection"
            connectionString="Data Source=(localdb)\MSSQLLocalDB;Initial
Catalog=DemoDB_Connection_01;Integrated Security=True"
            providerName="System.Data.SqlClient" />
    </connectionStrings>
    <system.web>
        <compilation debug="true" targetFramework="4.7.2" />
        <httpRuntime targetFramework="4.7.2" />
        <pages>
            <namespaces>
                <add namespace="System.Web.Optimization" />
            </namespaces>
            <controls>
                <add assembly="Microsoft.AspNet.Web.Optimization.WebForms"
namespace="Microsoft.AspNet.Web.Optimization.WebForms"
tagPrefix="webopt" />
            </controls>
        </pages>
    </system.web>
    <runtime>
        <assemblyBinding xmlns="urn:schemas-microsoft-com:asm.v1">
            <dependentAssembly>
                <assemblyIdentity name="Antlr3.Runtime"
publicKeyToken="eb42632606e9261f" />
                <bindingRedirect oldVersion="0.0.0.0-3.5.0.2" newVersion="3.5.0.2" />
            </dependentAssembly>
            <dependentAssembly>
                <assemblyIdentity name="Microsoft.Web.Infrastructure"
publicKeyToken="31bf3856ad364e35" />
                <bindingRedirect oldVersion="0.0.0.0-2.0.0.0" newVersion="2.0.0.0" />
            </dependentAssembly>
        </assemblyBinding>
    </runtime>
</configuration>
```

```

    <assemblyIdentity name="Newtonsoft.Json"
publicKeyToken="30ad4fe6b2a6aeed" />
    <bindingRedirect oldVersion="0.0.0.0-13.0.0.0" newVersion="13.0.0.0" />
  </dependentAssembly>
</dependentAssembly>
  <assemblyIdentity name="WebGrease"
publicKeyToken="31bf3856ad364e35" />
  <bindingRedirect oldVersion="0.0.0.0-1.6.5135.21930"
newVersion="1.6.5135.21930" />
</dependentAssembly>
</assemblyBinding>
</runtime>
<system.codedom>
  <compilers>
    <compiler language="c#;cs;csharp" extension=".cs"
type="Microsoft.CodeDom.Providers.DotNetCompilerPlatform.CSharpCodeProvi
der, Microsoft.CodeDom.Providers.DotNetCompilerPlatform, Version=2.0.1.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35" warningLevel="4"
compilerOptions="/langversion:default /nowarn:1659;1699;1701" />
    <compiler language="vb;vbs;visualbasic;vbscript" extension=".vb"
type="Microsoft.CodeDom.Providers.DotNetCompilerPlatform.VBCodeProvider,
Microsoft.CodeDom.Providers.DotNetCompilerPlatform, Version=2.0.1.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35" warningLevel="4"
compilerOptions="/langversion:default /nowarn:41008
/define:_MYTYPE=\&quot;Web\&quot; /optionInfer+" />
  </compilers>
</system.codedom>
</configuration>

```

## OUTPUT





## Practical 4

**Aim : Design a webpage to demonstrate a disconnected architecture.**

### Webform1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeBehind="WebForm1.aspx.cs" Inherits="WebApplication7.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <h2>Employee Records (Disconnected Architecture)</h2>

        <asp:GridView ID="GridView1" runat="server"
AutoGenerateColumns="true"></asp:GridView>

        <br />

        <asp:Button ID="btnLoad" runat="server" Text="Load Data"
OnClick="btnLoad_Click" />
        <asp:Button ID="btnUpdate" runat="server" Text="Update Data"
OnClick="btnUpdate_Click" />
    </form>
</body>
</html>
```

### Webform1.aspx.cs

```
using System;
using System.Data;
using System.Data.SqlClient;

namespace WebApplication7
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        // Only DataSet in ViewState (SAFE)
        private DataSet ds
        {
            get { return ViewState["ds"] as DataSet; }
            set { ViewState["ds"] = value; }
        }

        string connStr = @"Data Source=(localdb)\MSSQLLocalDB;Initial
Catalog=Employees;Integrated Security=True";

        protected void btnLoad_Click(object sender, EventArgs e)
        {
```

```

        using (SqlConnection con = new SqlConnection(connStr))
        {
            SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM
Employees", con);

            DataSet temp = new DataSet();
            da.Fill(temp, "Employees");

            ds = temp; // store ONLY dataset

            GridView1.DataSource = ds.Tables["Employees"];
            GridView1.DataBind();
        }
    }

    protected void btnUpdate_Click(object sender, EventArgs e)
    {
        if (ds == null)
        {
            Response.Write("Click Load first!");
            return;
        }

        using (SqlConnection con = new SqlConnection(connStr))
        {
            SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM
Employees", con);

            SqlCommandBuilder cb = new SqlCommandBuilder(da);

            // Modify data in memory
            if (ds.Tables["Employees"].Rows.Count > 0)
            {
                ds.Tables["Employees"].Rows[0]["Name"] = "Updated Name";
            }

            // Push update to DB
            da.Update(ds, "Employees");

            // Refresh Grid
            GridView1.DataSource = ds.Tables["Employees"];
            GridView1.DataBind();
        }
    }
}

```

## SQL

```

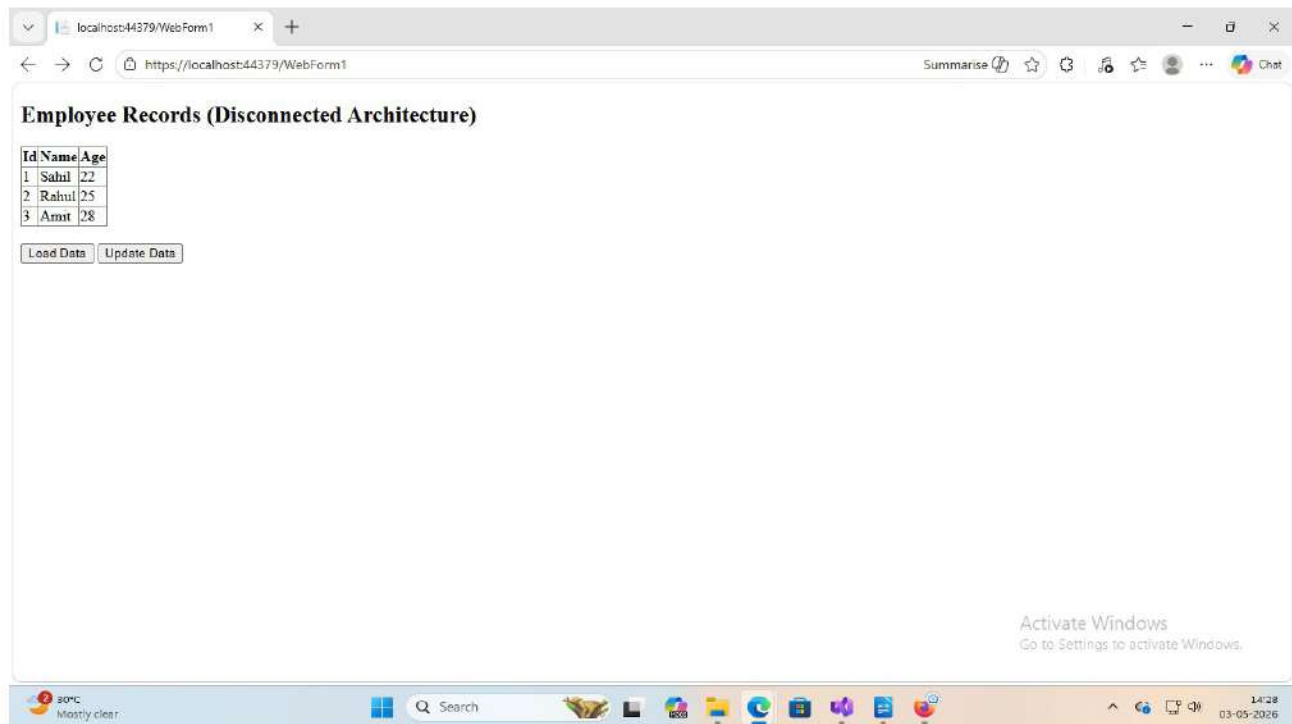
CREATE TABLE Employees (
    Id INT PRIMARY KEY IDENTITY(1,1),
    Name VARCHAR(50) NOT NULL,

```

```
Age INT NOT NULL
);

INSERT INTO Employees (Name, Age) VALUES
('Sahil', 22),
('Rahul', 25),
('Amit', 28);
```

## OUTPUT



The screenshot shows a web browser window with the address bar displaying 'localhost:44379/WebForm1'. The page title is 'Employee Records (Disconnected Architecture)'. Below the title, there is a table with the following data:

| Id | Name  | Age |
|----|-------|-----|
| 1  | Sahil | 22  |
| 2  | Rahul | 25  |
| 3  | Amit  | 28  |

Below the table, there are two buttons: 'Load Data' and 'Update Data'. At the bottom right of the page, there is a message: 'Activate Windows. Go to Settings to activate Windows.' The Windows taskbar is visible at the bottom, showing the date and time as 14:28 on 03-05-2026.



localhost:44379/WebForm1

Summarise ☆ ⚙️ 🔊 ☆ 👤 ... Chat

### Employee Records (Disconnected Architecture)

| Id | Name  | Age |
|----|-------|-----|
| 1  | Sahil | 22  |
| 2  | Rahul | 25  |
| 3  | Amit  | 28  |

Load DataUpdate Data

Activate Windows  
Go to Settings to activate Windows.

30°C  
Mostly clear

Search

14:29  
03-05-2026

## Practical 5

**Create a webpage that demonstrates the use of data bound controls of ASP.NET.**

Webform1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeBehind="WebForm1.aspx.cs" Inherits="charizard.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:DropDownList ID="DropDownList1" runat="server"
                AutoPostBack="True"
                OnSelectedIndexChanged="DropDownList1_SelectedIndexChanged">
</asp:DropDownList>
<br /><br />

            <asp:GridView ID="GridView1" runat="server"
                OnSelectedIndexChanged="GridView1_SelectedIndexChanged"></asp:GridVie
w>
<br /><br />

            <asp:DetailsView ID="DetailsView1" runat="server" Height="50px"
                Width="125px"></asp:DetailsView>
        </div>
    </form>
</body>
</html>
```

**Webform1.aspx.cs**

```
using System;
using System.Configuration;
using System.Data;
using System.Data.SqlClient;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace charizard
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        string cs =
            ConfigurationManager.ConnectionStrings["DemoConnection"].ConnectionString
        ;
    }
}
```

```

protected void Page_Load(object sender, EventArgs e)
{
    if (!IsPostBack)
    {
        BindGrid();
        BindDropdown();
    }
}

void BindGrid()
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM
Students", con);
        DataTable dt = new DataTable();
        da.Fill(dt);

        GridView1.DataSource = dt;
        GridView1.DataBind();

        // Show first record initially
        DetailsView1.DataSource = dt;
        DetailsView1.DataBind();
    }
}

void BindDropdown()
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM
Students", con);
        DataTable dt = new DataTable();
        da.Fill(dt);

        DropDownList1.DataSource = dt;
        DropDownList1.DataTextField = "Name";
        DropDownList1.DataValueField = "Id";
        DropDownList1.DataBind();
    }
}

// GridView selection → updates DetailsView
protected void GridView1_SelectedIndexChanged(object sender,
EventArgs e)
{
    int index = GridView1.SelectedIndex;

    if (index >= 0)
    {
        DetailsView1.PageIndex = index;
    }
}

```

```

    }
}

// 🔥 NEW: Dropdown selection → updates DetailsView
protected void DropDownList1_SelectedIndexChanged(object sender,
EventArgs e)
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        string query = "SELECT * FROM Students WHERE Id = @Id";
        SqlDataAdapter da = new SqlDataAdapter(query, con);
        da.SelectCommand.Parameters.AddWithValue("@Id",
DropDownList1.SelectedValue);

        DataTable dt = new DataTable();
        da.Fill(dt);

        DetailsView1.DataSource = dt;
        DetailsView1.DataBind();
    }
}
}
}
}

```

### SQL Query

```

CREATE TABLE Students (
    Id INT PRIMARY KEY IDENTITY,
    Name NVARCHAR(50),
    Age INT,
    Course NVARCHAR(50)
);

INSERT INTO Students (Name, Age, Course) VALUES
('Rahul', 20, 'BCA'),
('Anita', 22, 'MCA'),
('Vikas', 21, 'BSc');

```

### Output



localhost:44353/WebForm1

https://localhost:44353/WebForm1

Summarise

Chat

Rahul

| Id | Name  | Age | Course |
|----|-------|-----|--------|
| 1  | Rahul | 20  | BCA    |
| 2  | Anita | 22  | MCA    |
| 3  | Vikas | 21  | BSc    |

|        |       |
|--------|-------|
| Id     | 1     |
| Name   | Rahul |
| Age    | 20    |
| Course | BCA   |

29°C Clear

Search

02:08

28-04-2026

## Practical 6

**Design a webpage to demonstrate the working of a simple stored procedure.**

### Webform1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeBehind="WebForm1.aspx.cs" Inherits="WebApplication6.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <div>
                <h2>Stored Procedure Demo</h2>

                Enter Name:
                <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
                <br /><br />

                <asp:Button ID="btnSave" runat="server" Text="Save"
OnClick="btnSave_Click" />
                <br /><br />

                <asp:GridView ID="GridView1" runat="server"></asp:GridView>
            </div>
        </div>
    </form>
</body>
</html>
```

### Webform1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Configuration;
using System.Data;
using System.Data.SqlClient;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace WebApplication6
{
    public partial class WebForm1 : System.Web.UI.Page
    {

```

```

string cs =
ConfigurationManager.ConnectionStrings["conStr"].ConnectionString;

protected void Page_Load(object sender, EventArgs e)
{
    if (!IsPostBack)
    {
        LoadData();
    }
}

protected void btnSave_Click(object sender, EventArgs e)
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        SqlCommand cmd = new SqlCommand("InsertStudent", con);
        cmd.CommandType = CommandType.StoredProcedure;

        cmd.Parameters.AddWithValue("@Name", txtName.Text);

        con.Open();
        cmd.ExecuteNonQuery();
        con.Close();
    }

    txtName.Text = "";
    LoadData();
}

void LoadData()
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM
Students", con);
        DataTable dt = new DataTable();

        da.Fill(dt);
        GridView1.DataSource = dt;
        GridView1.DataBind();
    }
}
}
}

```

## SQL

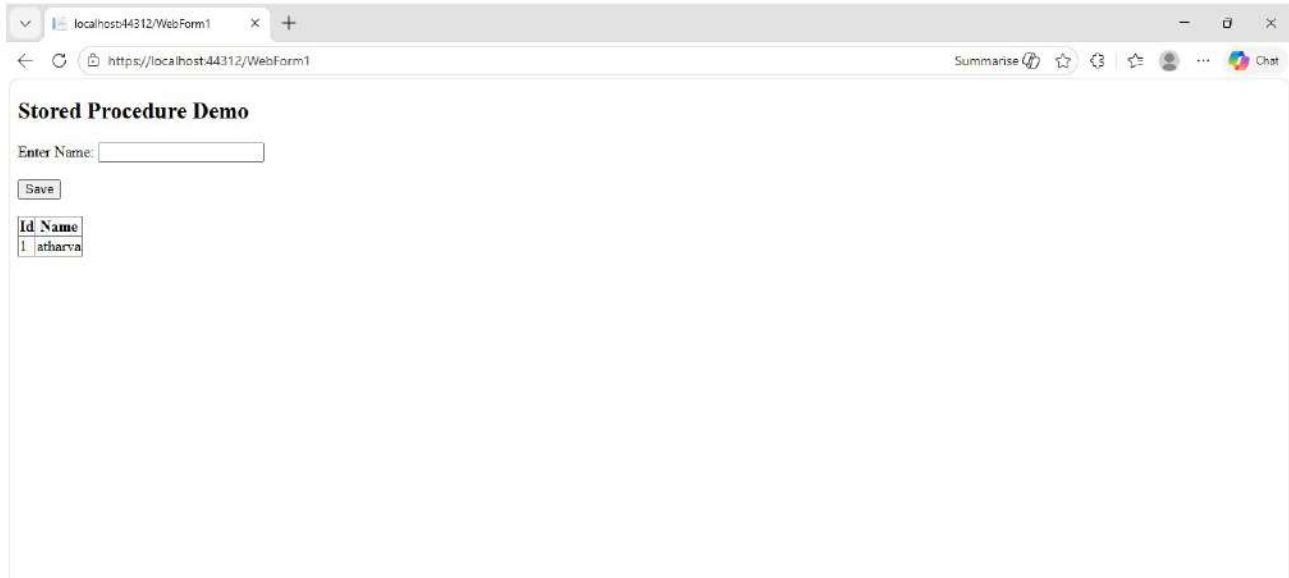
```

CREATE TABLE Students (
    Id INT PRIMARY KEY IDENTITY(1,1),
    Name NVARCHAR(50)
);

```

```
CREATE PROCEDURE InsertStudent
    @Name NVARCHAR(50)
AS
BEGIN
    INSERT INTO Students(@Name)
    VALUES(@Name)
END
```

## Output



Stored Procedure Demo

Enter Name:

| Id | Name    |
|----|---------|
| 1  | atharva |



## Practical 7

**Aim : Design a webpage to demonstrate the working of parameterized stored procedures.**

### WebForm1.aspx

```
using System;
using System.Collections.Generic;
using System.Configuration;
using System.Data;
using System.Data.SqlClient;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace WebApplication5
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        string cs =
ConfigurationManager.ConnectionStrings["DBCS"].ConnectionString;

        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                LoadData();
            }
        }

        protected void btnInsert_Click(object sender, EventArgs e)
        {
            using (SqlConnection con = new SqlConnection(cs))
            {
                SqlCommand cmd = new SqlCommand("InsertStudent", con);
                cmd.CommandType = CommandType.StoredProcedure;

                cmd.Parameters.AddWithValue("@Name", txtName.Text);
                cmd.Parameters.AddWithValue("@Marks", txtMarks.Text);

                con.Open();
                cmd.ExecuteNonQuery();
                con.Close();
            }

            LoadData();
        }

        private void LoadData()
```

```

    {
        using (SqlConnection con = new SqlConnection(cs))
        {
            SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM
Students1", con);
            DataTable dt = new DataTable();
            da.Fill(dt);

            GridView1.DataSource = dt;
            GridView1.DataBind();
        }
    }
}

```

### Webform1.aspx.cs

```

using System;
using System.Collections.Generic;
using System.Configuration;
using System.Data;
using System.Data.SqlClient;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

```

```

namespace WebApplication5

```

```

{
    public partial class WebForm1 : System.Web.UI.Page
    {
        string cs =
ConfigurationManager.ConnectionStrings["DBCS"].ConnectionString;

        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                LoadData();
            }
        }

        protected void btnInsert_Click(object sender, EventArgs e)
        {
            using (SqlConnection con = new SqlConnection(cs))
            {
                SqlCommand cmd = new SqlCommand("InsertStudent", con);
                cmd.CommandType = CommandType.StoredProcedure;

                cmd.Parameters.AddWithValue("@Name", txtName.Text);
            }
        }
    }
}

```

```

        cmd.Parameters.AddWithValue("@Marks", txtMarks.Text);

        con.Open();
        cmd.ExecuteNonQuery();
        con.Close();
    }

    LoadData();
}

private void LoadData()
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM
Students1", con);
        DataTable dt = new DataTable();
        da.Fill(dt);

        GridView1.DataSource = dt;
        GridView1.DataBind();
    }
}
}
}

```

### Web.config

```

<?xml version="1.0" encoding="utf-8"?>
<!--
    For more information on how to configure your ASP.NET application, please
    visit
    https://go.microsoft.com/fwlink/?LinkId=169433
-->
<configuration>
    <connectionStrings>
        <add name="DBCS"
            connectionString="Data Source=(localdb)\MSSQLLocalDB;Initial
Catalog=StudentDB;Integrated Security=True"
            providerName="System.Data.SqlClient"/>
    </connectionStrings>
    <system.web>
        <compilation debug="true" targetFramework="4.7.2" />
        <httpRuntime targetFramework="4.7.2" />
        <pages>
            <namespaces>
                <add namespace="System.Web.Optimization" />
            </namespaces>
            <controls>

```



```

        <add assembly="Microsoft.AspNet.Web.Optimization.WebForms"
namespace="Microsoft.AspNet.Web.Optimization.WebForms"
tagPrefix="webopt" />
    </controls>
</pages>
</system.web>
<runtime>
    <assemblyBinding xmlns="urn:schemas-microsoft-com:asm.v1">
        <dependentAssembly>
            <assemblyIdentity name="Antlr3.Runtime"
publicKeyToken="eb42632606e9261f" />
            <bindingRedirect oldVersion="0.0.0.0-3.5.0.2" newVersion="3.5.0.2" />
        </dependentAssembly>
        <dependentAssembly>
            <assemblyIdentity name="Microsoft.Web.Infrastructure"
publicKeyToken="31bf3856ad364e35" />
            <bindingRedirect oldVersion="0.0.0.0-2.0.0.0" newVersion="2.0.0.0" />
        </dependentAssembly>
        <dependentAssembly>
            <assemblyIdentity name="Newtonsoft.Json"
publicKeyToken="30ad4fe6b2a6aeed" />
            <bindingRedirect oldVersion="0.0.0.0-13.0.0.0" newVersion="13.0.0.0" />
        </dependentAssembly>
        <dependentAssembly>
            <assemblyIdentity name="WebGrease"
publicKeyToken="31bf3856ad364e35" />
            <bindingRedirect oldVersion="0.0.0.0-1.6.5135.21930"
newVersion="1.6.5135.21930" />
        </dependentAssembly>
    </assemblyBinding>
</runtime>
<system.codedom>
    <compilers>
        <compiler language="c#;cs;csharp" extension=".cs"
type="Microsoft.CodeDom.Providers.DotNetCompilerPlatform.CSharpCodeProvi
der, Microsoft.CodeDom.Providers.DotNetCompilerPlatform, Version=2.0.1.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35" warningLevel="4"
compilerOptions="/langversion:default /nowarn:1659;1699;1701" />
        <compiler language="vb;vbs;visualbasic;vbscript" extension=".vb"
type="Microsoft.CodeDom.Providers.DotNetCompilerPlatform.VBCodeProvider,
Microsoft.CodeDom.Providers.DotNetCompilerPlatform, Version=2.0.1.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35" warningLevel="4"
compilerOptions="/langversion:default /nowarn:41008
/define:_MYTYPE=\"Web\" /optionInfer+" />
    </compilers>
</system.codedom>
</configuration>

```

## SQL

```

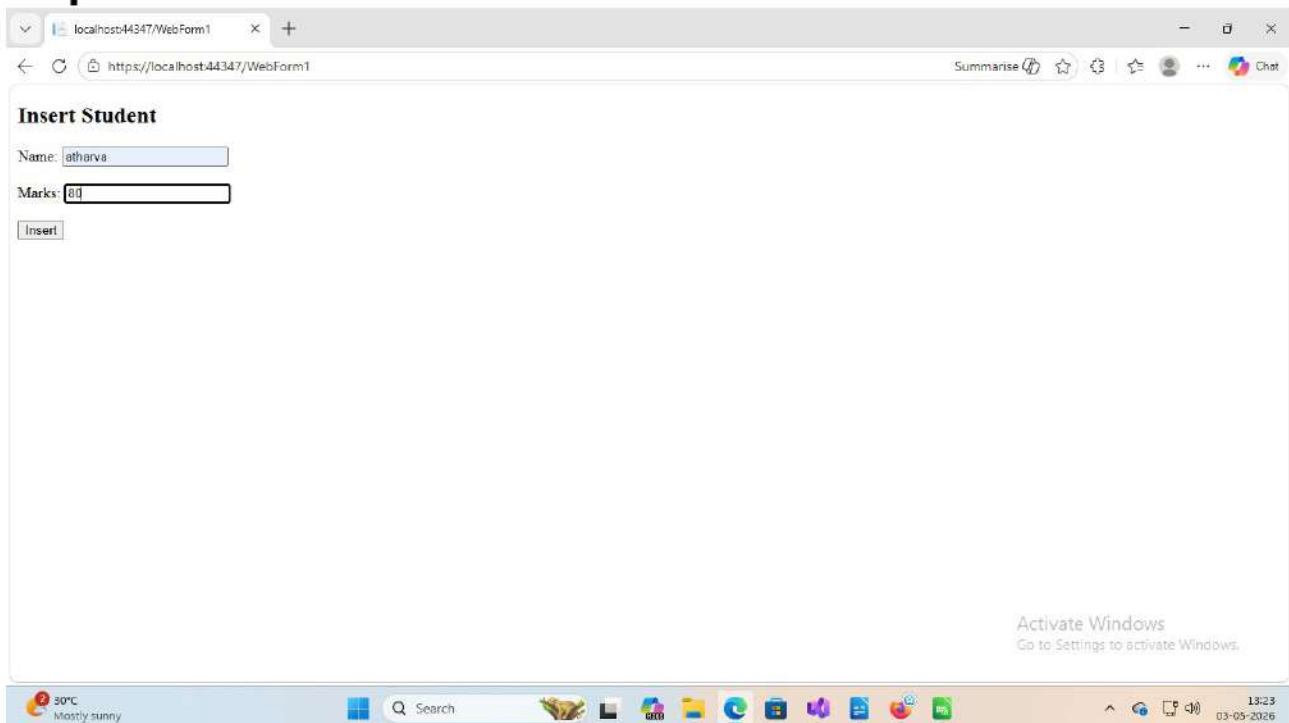
CREATE TABLE Students1 (
    Id INT PRIMARY KEY IDENTITY(1,1),

```

```
Name NVARCHAR(50),
Marks INT
);

CREATE PROCEDURE InsertStudent
    @Name NVARCHAR(50),
    @Marks INT
AS
BEGIN
    INSERT INTO Students1 (Name, Marks)
    VALUES (@Name, @Marks)
END;
```

## Output



The screenshot shows a web browser window with the address bar displaying 'localhost:44347/WebForm1'. The page title is 'Insert Student'. The form contains two input fields: 'Name' with the value 'atharva' and 'Marks' with the value '80'. Below the fields is an 'Insert' button. The browser's address bar shows the URL 'https://localhost:44347/WebForm1'. The Windows taskbar at the bottom shows the date and time as '13:23 03-05-2026' and the weather as '30°C Mostly sunny'.

**Insert Student**

Name:

Marks:

Activate Windows  
Go to Settings to activate Windows.



localhost:44347/WebForm1

https://localhost:44347/WebForm1

Summarise

Chat

### Insert Student

Name:

Marks:

| Id | Name    | Marks |
|----|---------|-------|
| 1  | atharva | 80    |

Activate Windows

Go to Settings to activate Windows.

Hot days ahead  
30°C

Search

13:23

03-05-2026

## Practical 8

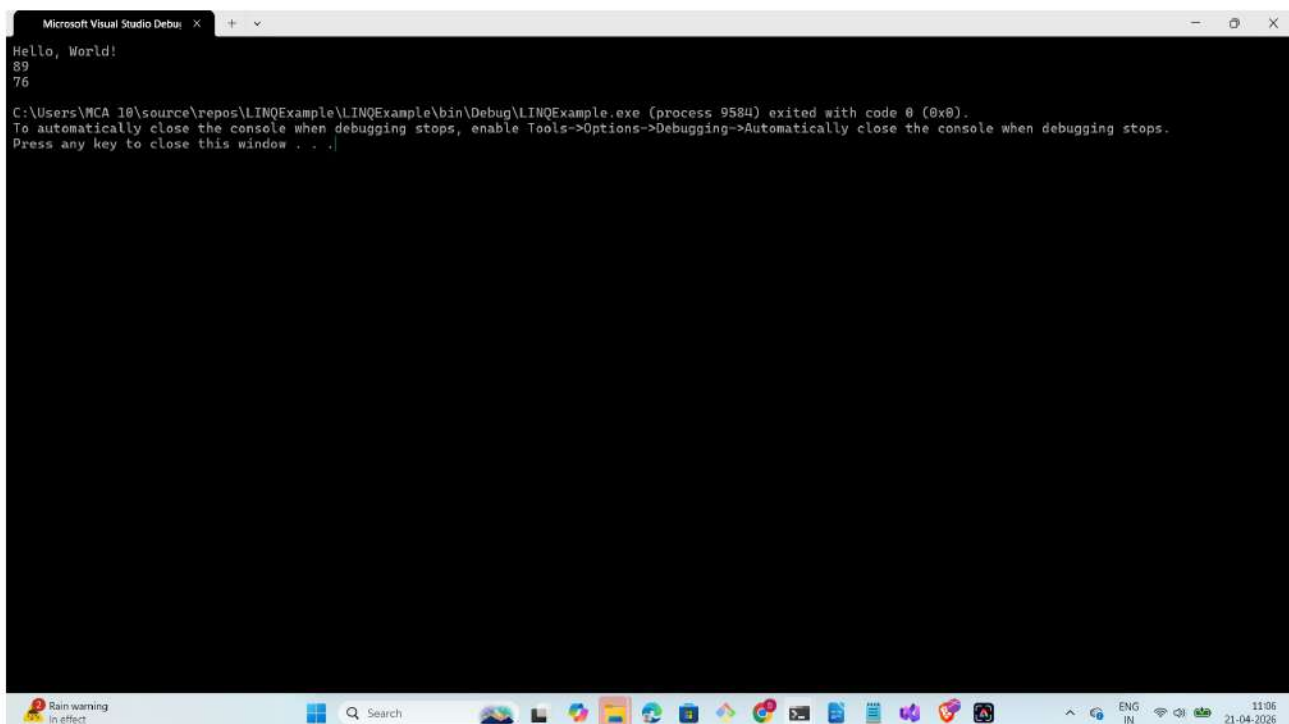
**Aim : To design a webpage to display the use of LINQ.**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace LINQExample
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Hello, World!");
            int[] numbers = { 1, 2, 3, 4, 5, 30, 89, 76 };
            var result = from n in numbers where n > 30 select n;

            foreach (var item in result)
            {
                Console.WriteLine(item);
            }
        }
    }
}
```

## OUTPUT



The screenshot shows a Windows desktop with a taskbar at the bottom. The taskbar includes a weather widget for 'Rain warning in effect', a search bar, and several application icons. The main focus is a 'Microsoft Visual Studio Debug' console window. The console has a black background with white text. It displays the output 'Hello, World!' followed by the numbers '89' and '76' on separate lines. Below the output, a message states: 'C:\Users\WCA 10\source\repos\LINQExample\LINQExample\bin\Debug\LINQExample.exe (process 9584) exited with code 0 (0x0). To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops. Press any key to close this window . . .'. The console window has standard Windows window controls (minimize, maximize, close) in the top right corner.

## Practical 9

**Aim : Build websites to demonstrate the working of entity frameworks in dot net.**

### Create.cshtml

@model WebApplication4.Models.Student

<h2>Add Student</h2>

@using (Html.BeginForm())

{

<label>Name</label>

@Html.TextBoxFor(m => m.Name)

<br />

<label>Age</label>

@Html.TextBoxFor(m => m.Age)

<br />

<label>Course</label>

@Html.TextBoxFor(m => m.Course)

<br />

<input type="submit" value="Save" />

}

### Index.cshtml

@model IEnumerable<WebApplication4.Models.Student>

<h2>Student List</h2>

<a href="/Student/Create">Add New</a>

<table border="1">

<tr>

<th>Name</th>

<th>Age</th>

<th>Course</th>

</tr>

@foreach (var item in Model)

{

<tr>

<td>@item.Name</td>

<td>@item.Age</td>

<td>@item.Course</td>

</tr>

```
}
```

```
</table>
```

### **StudentController.cs**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Mvc;
using WebApplication4.Models;

namespace WebApplication4.Controllers
{
    public class StudentController : Controller
    {
        private StudentContext db = new StudentContext();

        // READ
        public ActionResult Index()
        {
            return View(db.Students.ToList());
        }

        // CREATE
        public ActionResult Create()
        {
            return View();
        }

        [HttpPost]
        public ActionResult Create(Student s)
        {
            if (ModelState.IsValid)
            {
                db.Students.Add(s);
                db.SaveChanges();
                return RedirectToAction("Index");
            }
            return View(s);
        }
    }
}
```

### **StudentContext.cs**

```
using System;
using System.Collections.Generic;
using System.Data.Entity;
using System.Linq;
using System.Web;
```



```

using WebApplication4.Models;
using System.Data.Entity;

namespace WebApplication4.Models
{
    public class StudentContext : DbContext
    {
        public StudentContext() : base("StudentContext")
        {
        }

        public DbSet<Student> Students { get; set; }
    }
}

```

### Student.cs

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.ComponentModel.DataAnnotations;

namespace WebApplication4.Models
{
    public class Student
    {
        public int Id { get; set; }

        [Required]
        public string Name { get; set; }

        public int Age { get; set; }

        public string Course { get; set; }
    }
}

```

### Web.Config

```

<?xml version="1.0" encoding="utf-8"?>
<configuration>

    <!-- MUST BE FIRST -->
    <configSections>
        <section name="entityFramework"

type="System.Data.Entity.Internal.ConfigFile.EntityFrameworkSection,
EntityFramework, Version=6.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089"
        requirePermission="false" />
    </configSections>

```

```

<connectionStrings>
  <add name="StudentContext"
    connectionString="Data Source=(localdb)\MSSQLLocalDB;Initial
Catalog=EFStudentDB;Integrated Security=True"
    providerName="System.Data.SqlClient" />
</connectionStrings>

<appSettings>
  <add key="webpages:Version" value="3.0.0.0" />
  <add key="webpages:Enabled" value="false" />
  <add key="ClientValidationEnabled" value="true" />
  <add key="UnobtrusiveJavaScriptEnabled" value="true" />
</appSettings>

<system.web>
  <compilation debug="true" targetFramework="4.7.2" />
  <httpRuntime targetFramework="4.7.2" />
</system.web>

<runtime>
  <assemblyBinding xmlns="urn:schemas-microsoft-com:asm.v1">

    <dependentAssembly>
      <assemblyIdentity name="Antlr3.Runtime"
publicKeyToken="eb42632606e9261f" />
      <bindingRedirect oldVersion="0.0.0.0-3.5.0.2"
newVersion="3.5.0.2" />
    </dependentAssembly>

    <dependentAssembly>
      <assemblyIdentity name="Microsoft.Web.Infrastructure"
publicKeyToken="31bf3856ad364e35" />
      <bindingRedirect oldVersion="0.0.0.0-2.0.0.0"
newVersion="2.0.0.0" />
    </dependentAssembly>

    <dependentAssembly>
      <assemblyIdentity name="Newtonsoft.Json"
publicKeyToken="30ad4fe6b2a6aeed" />
      <bindingRedirect oldVersion="0.0.0.0-13.0.0.0"
newVersion="13.0.0.0" />
    </dependentAssembly>

    <dependentAssembly>
      <assemblyIdentity name="System.Web.Optimization"
publicKeyToken="31bf3856ad364e35" />
      <bindingRedirect oldVersion="1.0.0.0-1.1.0.0"
newVersion="1.1.0.0" />
    </dependentAssembly>

    <dependentAssembly>

```



```

        <assemblyIdentity name="WebGreas"
publicKeyToken="31bf3856ad364e35" />
        <bindingRedirect oldVersion="0.0.0.0-1.6.5135.21930"
newVersion="1.6.5135.21930" />
    </dependentAssembly>

    <dependentAssembly>
        <assemblyIdentity name="System.Web.Helpers"
publicKeyToken="31bf3856ad364e35" />
        <bindingRedirect oldVersion="1.0.0.0-3.0.0.0"
newVersion="3.0.0.0" />
    </dependentAssembly>

    <dependentAssembly>
        <assemblyIdentity name="System.Web.WebPages"
publicKeyToken="31bf3856ad364e35" />
        <bindingRedirect oldVersion="1.0.0.0-3.0.0.0"
newVersion="3.0.0.0" />
    </dependentAssembly>

    <dependentAssembly>
        <assemblyIdentity name="System.Web.Mvc"
publicKeyToken="31bf3856ad364e35" />
        <bindingRedirect oldVersion="1.0.0.0-5.2.9.0"
newVersion="5.2.9.0" />
    </dependentAssembly>

</assemblyBinding>
</runtime>

<system.codedom>
    <compilers>
        <compiler language="c#;cs;csharp" extension=".cs"
type="Microsoft.CodeDom.Providers.DotNetCompilerPlatform.CSharpCodeProvi
der, Microsoft.CodeDom.Providers.DotNetCompilerPlatform, Version=2.0.1.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35"
        warningLevel="4"
        compilerOptions="/langversion:default
/nowarn:1659;1699;1701" />

        <compiler language="vb;vbs;visualbasic;vbscript"
extension=".vb"
type="Microsoft.CodeDom.Providers.DotNetCompilerPlatform.VBCodeProvider,
Microsoft.CodeDom.Providers.DotNetCompilerPlatform, Version=2.0.1.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35"
        warningLevel="4"
        compilerOptions="/langversion:default /nowarn:41008
/define:_MYTYPE="Web" /optionInfer+" />
    </compilers>
</system.codedom>

```

```

<entityFramework>
  <providers>
    <provider invariantName="System.Data.SqlClient"
type="System.Data.Entity.SqlServer.SqlProviderServices,
EntityFramework.SqlServer" />
  </providers>
</entityFramework>

</configuration>

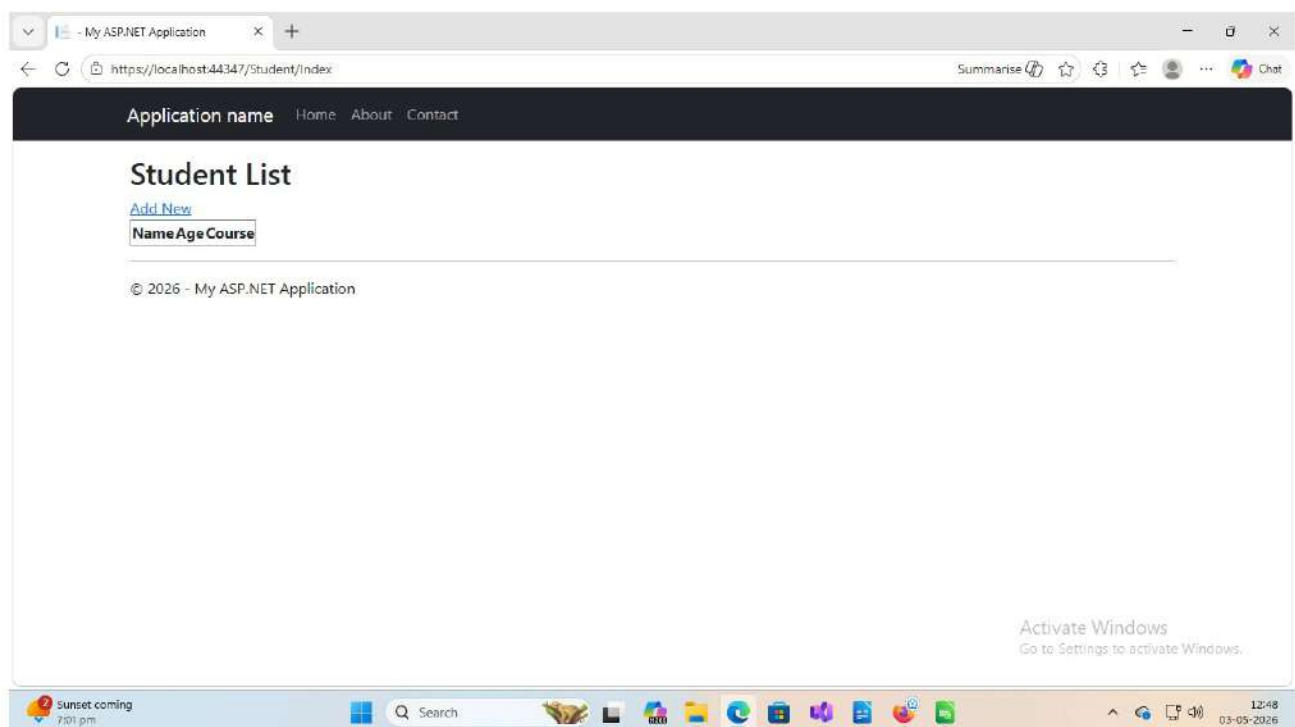
```

## SQL

```

CREATE TABLE Students (
  Id INT PRIMARY KEY IDENTITY,
  Name NVARCHAR(100),
  Age INT,
  Course NVARCHAR(100)
);

```







## Practical 10

### Aim : Design Web Applications using Client Side Session Management

#### Default.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs" Inherits="ClientSessionDemo.Default" %>
```

```
<!DOCTYPE html>
<html>
<head runat="server">
  <title>Login Page</title>
</head>
<body>
  <form id="form1" runat="server">
    <div>
      <h2>Enter Your Name</h2>
      <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
      <br /><br />
      <asp:Button ID="btnSubmit" runat="server" Text="Submit" On-
Click="btnSubmit_Click" />
    </div>
  </form>
</body>
</html>
```

#### Default.aspx.cs

```
using System;
using System.Web;
namespace ClientSessionDemo
{
  public partial class Default : System.Web.UI.Page
  {
    protected void btnSubmit_Click(object sender, EventArgs e)
    {
      // Create cookie
      HttpCookie userCookie = new HttpCookie("UserInfo");

      // Store value
      userCookie["Name"] = txtName.Text;

      // Set expiry time (1 day)
      userCookie.Expires = DateTime.Now.AddDays(1);

      // Add cookie to response
      Response.Cookies.Add(userCookie);

      // Redirect to next page
      Response.Redirect("Welcome.aspx");
    }
  }
}
```

## Welcome.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Welcome.aspx.cs" Inherits="ClientSessionDemo.Welcome" %>
```

```
<!DOCTYPE html>
<html>
<head runat="server">
  <title>Welcome Page</title>
</head>
<body>
  <form id="form1" runat="server">
    <div>
      <h2>Welcome Page</h2>
      <asp:Label ID="lblMessage" runat="server"></asp:Label>
    </div>
  </form>
</body>
</html>
```

## Welcome.aspx.cs

```
using System;
namespace ClientSessionDemo
{
    public partial class Welcome : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (Request.Cookies["UserInfo"] != null)
            {
                string name = Request.Cookies["UserInfo"]["Name"];
                lblMessage.Text = "Hello, " + name + "!";
            }
            else
            {
                lblMessage.Text = "No cookie found!";
            }
        }
    }
}
```

## OUTPUT

Login Page

localhost:50568/Default

☆

Enter Your Name

Student

Submit



## Practical 11

### Aim : Design Web Applications using Server Side Session Management Techniques

#### 1) Default.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs" Inherits="ServerSessionDemo.Default" %>
<!DOCTYPE html>
<html>
<head runat="server">
  <title>Login Page</title>
</head>
<body>
  <form id="form1" runat="server">
    <div>
      <h2>Enter Your Name</h2>
      <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
      <br /><br />
      <asp:Button ID="btnSubmit" runat="server" Text="Submit" On-
Click="btnSubmit_Click" />
    </div>
  </form>
</body>
</html>
```

#### Default.aspx.cs

```
using System;
namespace ServerSessionDemo
{
  public partial class Default : System.Web.UI.Page
  {
    protected void btnSubmit_Click(object sender, EventArgs e)
    {
      // Store data in Session
      Session["UserName"] = txtName.Text;
      // Redirect to next page
      Response.Redirect("Welcome.aspx");
    }
  }
}
```

#### Welcome.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Welcome.aspx.cs" Inherits="ServerSessionDemo.Welcome" %>
<!DOCTYPE html>
<html>
<head runat="server">
  <title>Welcome Page</title>
</head>
<body>
  <form id="form1" runat="server">
```

```

        <div>
            <h2>Welcome Page</h2>
            <asp:Label ID="lblMessage" runat="server"></asp:Label>
        </div>
    </form>
</body>
</html>

```

#### 4) Welcome.aspx.cs

```
using System;
```

```
namespace ServerSessionDemo
```

```

{
    public partial class Welcome : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (Session["UserName"] != null)
            {
                lblMessage.Text = "Hello, " + Session["UserName"].ToString();
            }
            else
            {
                lblMessage.Text = "Session expired or not set!";
            }
        }
    }
}

```

#### OUTPUT



Login Page

localhost:50763/Default

Enter Your Name

Submit

## Practical 12

### Build a web page using AJAX Controls.

#### Webform1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeBehind="WebForm2.aspx.cs" Inherits="pokemon.WebForm2" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>AJAX Demo</title>
</head>
<body>
    <form id="form1" runat="server">

        <!-- Required for AJAX -->
        <asp:ScriptManager ID="ScriptManager1" runat="server" />

        <div style="margin:50px">

            <h2>ASP.NET AJAX Example</h2>

            <!-- UpdatePanel for Partial Page Update -->
            <asp:UpdatePanel ID="UpdatePanel1" runat="server">
                <ContentTemplate>

                    <asp:Label ID="lblTime" runat="server" Text="Click button to get
time" Font-Size="Large"></asp:Label>
                    <br /><br />

                    <asp:Button ID="btnGetTime" runat="server" Text="Get Current
Time" OnClick="btnGetTime_Click" />

                </ContentTemplate>
            </asp:UpdatePanel>

        </div>
    </form>
</body>
</html>
```

#### Webform2.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace pokemon
{
    public partial class WebForm2 : System.Web.UI.Page
```

```
{
    protected void Page_Load(object sender, EventArgs e)
    {

    }

    protected void btnGetTime_Click(object sender, EventArgs e)
    {
        lblTime.Text = "Current Time: " + DateTime.Now.ToString();
    }
}
}
```

## OUTPUT





## Practical 13

**Aim : Design Web Application to produce and Consume a web Service**

### Webform1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeBehind="WebForm1.aspx.cs" Inherits="WebApplication10.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div><asp:TextBox ID="txtId" runat="server"></asp:TextBox>
<asp:Button ID="btnGet" runat="server" Text="Get Student Name"
OnClick="btnGet_Click" />
<br /><br />
<asp:Label ID="lblResult" runat="server"></asp:Label>
        </div>
    </form>
</body>
</html>
```

### Webform1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace WebApplication10
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void btnGet_Click(object sender, EventArgs e)
        {
            int id;

            if (int.TryParse(txtId.Text, out id))
            {
                ServiceReference1.StudentServiceSoapClient obj =
                    new ServiceReference1.StudentServiceSoapClient();

                string name = obj.GetStudentName(id);

                lblResult.Text = "Student Name: " + name;
            }
        }
    }
}
```

```

        else
        {
            lblResult.Text = "✗ Please enter a valid numeric ID!";
        }
    }
}

```

### **StudentServices.asmx.cs**

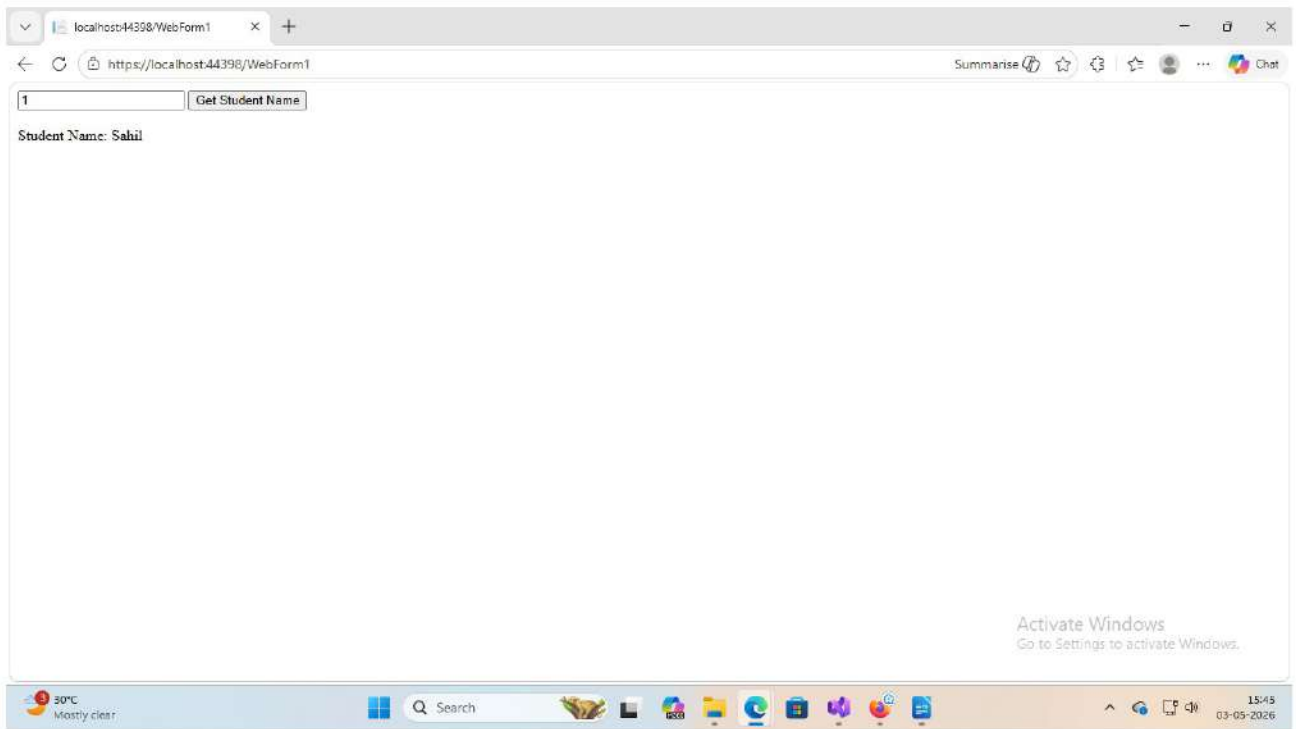
```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Services;

namespace WebApplication9
{
    [WebService(Namespace = "http://tempuri.org/")]
    public class StudentService : WebService
    {
        [WebMethod]
        public string GetStudentName(int id)
        {
            if (id == 1)
                return "Sahil";
            else if (id == 2)
                return "Rahul";
            else
                return "Unknown Student";
        }
    }
}

```

### **OUTPUT**



## Practical 14

**Aim : Design web application using MVC framework**

**Index.chtml**

```
@model List<MVCStudentApp.Models.Student>

<!DOCTYPE html>
<html>
<head>
    <title>Student Dashboard</title>

    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

    <style>
        body {
            font-family: Arial;
            margin: 20px;
            background: #f4f6f9;
            text-align: center;
        }

        .card {
            background: white;
            padding: 20px;
            border-radius: 15px;
            width: 400px;
            margin: auto;
            box-shadow: 0 5px 15px rgba(0,0,0,0.1);
        }

        canvas {
            margin-top: 20px;
        }

        input, button {
            padding: 8px;
            margin: 5px;
        }
    </style>
</head>

<body>

    <h2> 📊 Student Dashboard</h2>

    <div class="card">

        <h3>Total Students: @ViewBag.TotalStudents</h3>

        <form method="post" action="/Student/AddStudent">
```



```

        <input type="text" name="name" placeholder="Enter student
name" />
        <button type="submit">Add</button>
    </form>

    <canvas id="studentChart"></canvas>

    <ul>
        @foreach (var student in Model)
        {
            <li>@student.Name</li>
        }
    </ul>

</div>

<script>
var total = @ViewBag.TotalStudents;

var ctx = document.getElementById('studentChart').getContext('2d');

new Chart(ctx, {
    type: 'bar',
    data: {
        labels: ['Students'],
        datasets: [{
            label: 'Total Count',
            data: [total]
        }]
    }
});
</script>

</body>
</html>

```

### **Student Controller.cs**

```

using System.Collections.Generic;
using System.Web.Mvc;
using MVCStudentApp.Models;

namespace MVCStudentApp.Controllers
{
    public class StudentController : Controller
    {
        static List<Student> students = new List<Student>();

        public ActionResult Index()
        {
            ViewBag.TotalStudents = students.Count;

```

```

        return View(students);
    }

    [HttpPost]
    public ActionResult AddStudent(string name)
    {
        if (!string.IsNullOrEmpty(name))
        {
            students.Add(new Student { Name = name });
        }
        return RedirectToAction("Index");
    }
}
}

```

### Student.cs

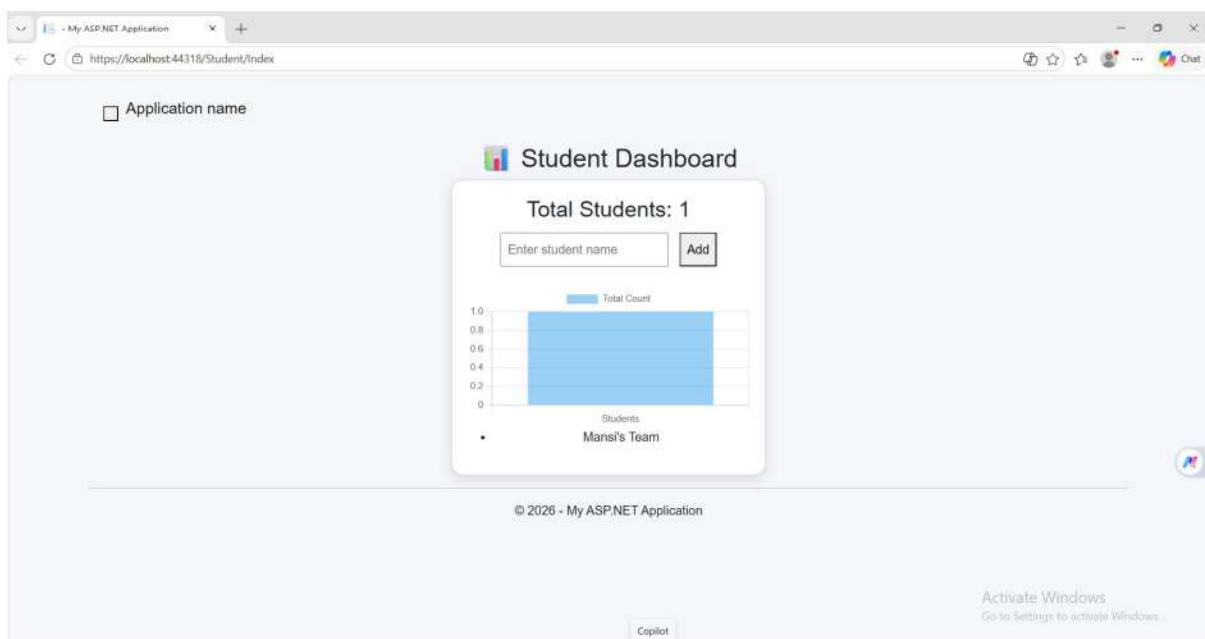
```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;

namespace MVCStudentApp.Models
{
    public class Student
    {
        public string Name { get; set; }
    }
}

```

## OUTPUT



## **Practical 15**

**Aim : Design MVC application using entity framework**

### **Install Entity Framework**

Go to:

Tools → NuGet Package Manager → Package Manager Console

Run:

Install-Package EntityFramework

### **1) Models/Student.cs**

using System.Data.Entity;

```
namespace EFStudentApp.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public int Marks { get; set; }
    }

    public class StudentContext : DbContext
    {
        public StudentContext() : base("StudentDB")
        {
        }

        public DbSet<Student> Students { get; set; }
    }
}
```

### **2) Controllers/StudentController.cs**

```
using System.Linq;
using System.Web.Mvc;
using EFStudentApp.Models;

namespace EFStudentApp.Controllers
{
    public class StudentController : Controller
    {
        StudentContext db = new StudentContext();

        public ActionResult Index()
```

```

    {
        var data = db.Students.ToList();
        return View(data);
    }

    public ActionResult Create()
    {
        return View();
    }

    [HttpPost]
    public ActionResult Create(Student s)
    {
        db.Students.Add(s);
        db.SaveChanges();
        return RedirectToAction("Index");
    }
}

```

### 3) Views/Student/Create.cshtml

```

@model EFStudentApp.Models.Student

<h2>Add Student</h2>

@using (Html.BeginForm())
{
    <div>
        Name: @Html.TextBoxFor(m => m.Name)
    </div>

    <br />

    <div>
        Marks: @Html.TextBoxFor(m => m.Marks)
    </div>

    <br />

    <input type="submit" value="Save" />
}

```

### 4) Views/Student/Index.cshtml

```

@model List<EFStudentApp.Models.Student>

<h2>Student List</h2>

```



```
<a href="/Student/Create">Add Student</a>
```

```
<table border="1">
  <tr>
    <th>ID</th>
    <th>Name</th>
    <th>Marks</th>
  </tr>

  @foreach (var s in Model)
  {
    <tr>
      <td>@s.Id</td>
      <td>@s.Name</td>
      <td>@s.Marks</td>
    </tr>
  }
</table>
```

## 5) Web.config

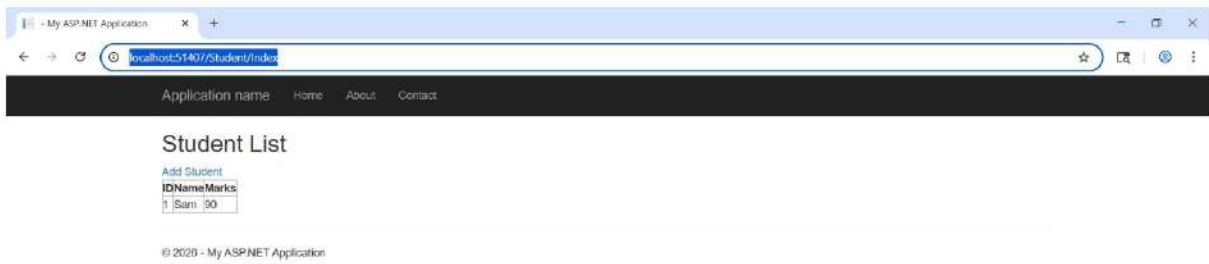
```
<connectionStrings>
  <add name="StudentDB"
    connectionString="Data Source=.;Initial Catalog=StudentDB;Integrated
Security=True"
    providerName="System.Data.SqlClient" />
</connectionStrings>
```

## 6) Enable Database Creation

Run this in Package Manager Console:

```
Enable-Migrations
Add-Migration InitialCreate
Update-Database
```

## Output



## Practical 16

### Design MVC Application to perform CRUD operation.

#### 1) Controller/EmployeeController.cs

```
using System.Linq;
using System.Web.Mvc;
using EmployeeCRUD.Models;

namespace EmployeeApp.Controllers
{
    public class EmployeeController : Controller
    {
        EmployeeContext db = new EmployeeContext();

        // READ (Display)
        public ActionResult Index()
        {
            return View(db.Employees.ToList());
        }

        // CREATE
        public ActionResult Create()
        {
            return View();
        }

        [HttpPost]
        public ActionResult Create(Employee e)
        {
            db.Employees.Add(e);
            db.SaveChanges();
            return RedirectToAction("Index");
        }

        // UPDATE
        public ActionResult Edit(int id)
        {
            var emp = db.Employees.Find(id);
            return View(emp);
        }

        [HttpPost]
        public ActionResult Edit(Employee e)
        {
            db.Entry(e).State = System.Data.Entity.EntityState.Modified;
            db.SaveChanges();
            return RedirectToAction("Index");
        }
    }
}
```

```

// DELETE
public ActionResult Delete(int id)
{
    var emp = db.Employees.Find(id);
    return View(emp);
}

[HttpPost, ActionName("Delete")]
public ActionResult DeleteConfirmed(int id)
{
    var emp = db.Employees.Find(id);
    db.Employees.Remove(emp);
    db.SaveChanges();
    return RedirectToAction("Index");
}
}
}

```

## 2) Models/Employee.cs

```

using System.Data.Entity;

namespace EmployeeCRUD.Models
{
    public class Employee
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Department { get; set; }
        public int Salary { get; set; }
    }

    public class EmployeeContext : DbContext
    {
        public EmployeeContext() : base("EmployeeDB")
        {
        }

        public DbSet<Employee> Employees { get; set; }
    }
}

```

## 3) Views/Employee/Create.cs

```

@model EmployeeCRUD.Models.Employee

<h2>Add Employee</h2>

@using (Html.BeginForm())

```

```

{
    <div>Name: @Html.TextBoxFor(m => m.Name)</div>
    <div>Department: @Html.TextBoxFor(m => m.Department)</div>
    <div>Salary: @Html.TextBoxFor(m => m.Salary)</div>

    <br />
    <input type="submit" value="Save" />
}

```

#### 4) Views/Employee/Delete.cs

```

@model EmployeeCRUD.Models.Employee

<h2>Delete Employee</h2>

<p>Are you sure you want to delete this?</p>

<div>
    Name: @Model.Name <br />
    Department: @Model.Department <br />
    Salary: @Model.Salary
</div>

@using (Html.BeginForm())
{
    @Html.HiddenFor(m => m.Id)
    <input type="submit" value="Delete" />
}

```

#### 5) Views/Employee/Details.cs

```

@model EmployeeCRUD.Models.Employee

@{
    ViewBag.Title = "Details";
}

<h2>Details</h2>

<div>
    <h4>Employee</h4>
    <hr />
    <dl class="dl-horizontal">
        <dt>
            @Html.DisplayNameFor(model => model.Name)
        </dt>

        <dd>
            @Html.DisplayFor(model => model.Name)
        </dd>
    </dl>

```



```

</dd>

<dt>
    @Html.DisplayNameFor(model => model.Department)
</dt>

<dd>
    @Html.DisplayFor(model => model.Department)
</dd>

<dt>
    @Html.DisplayNameFor(model => model.Salary)
</dt>

<dd>
    @Html.DisplayFor(model => model.Salary)
</dd>

</dl>
</div>
<p>
    @Html.ActionLink("Edit", "Edit", new { id = Model.Id }) |
    @Html.ActionLink("Back to List", "Index")
</p>

```

## 6) Views/Employee/Edit.cs

```

@model EmployeeCRUD.Models.Employee
<h2>Edit Employee</h2>

@using (Html.BeginForm())
{
    @Html.HiddenFor(m => m.Id)

    <div>Name: @Html.TextBoxFor(m => m.Name)</div>
    <div>Department: @Html.TextBoxFor(m => m.Department)</div>
    <div>Salary: @Html.TextBoxFor(m => m.Salary)</div>

    <br />
    <input type="submit" value="Update" />
}

```

## 7) Views/Employee/Index.cs

```

@model List<EmployeeCRUD.Models.Employee>
<h2>Employee List</h2>
<a href="/Employee/Create">Add Employee</a>
<table border="1">
    <tr>

```

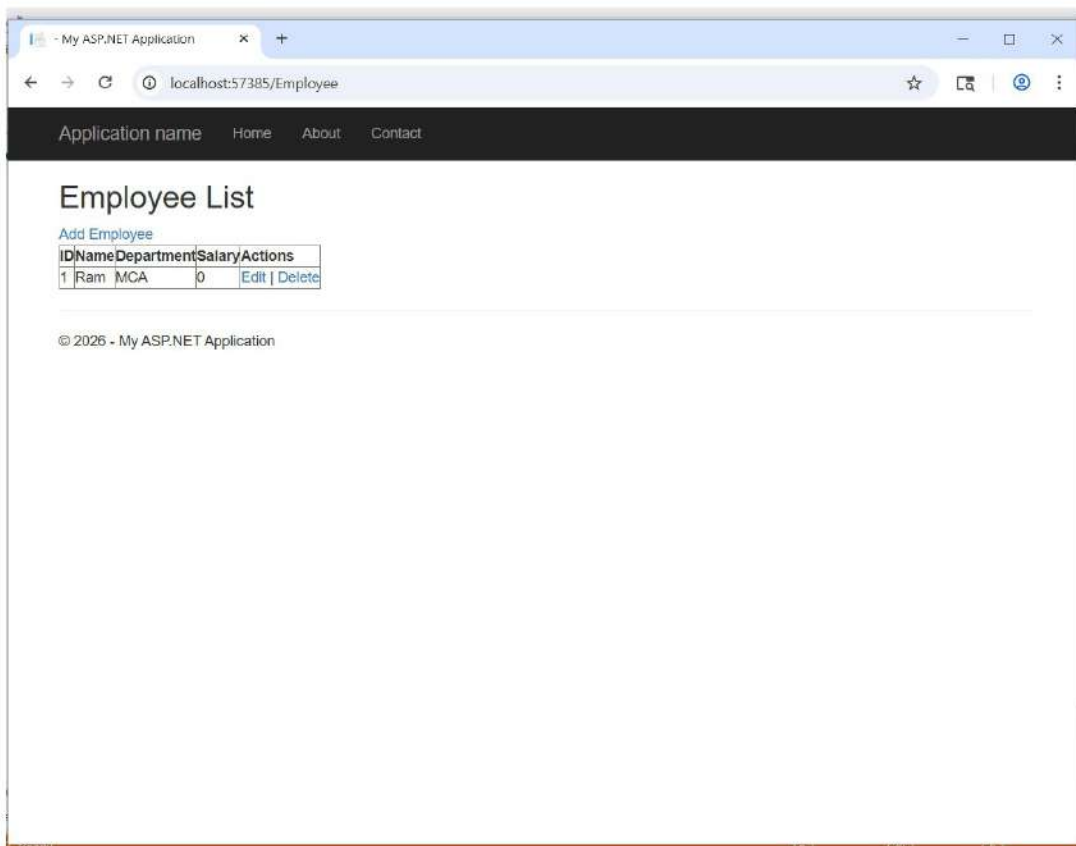
```

<th>ID</th>
<th>Name</th>
<th>Department</th>
<th>Salary</th>
<th>Actions</th>
</tr>

@foreach (var e in Model)
{
    <tr>
        <td>@e.Id</td>
        <td>@e.Name</td>
        <td>@e.Department</td>
        <td>@e.Salary</td>
        <td>
            <a href="@Url.Action("Edit", "Employee", new { id =
e.Id })">Edit</a> |
            <a href="@Url.Action("Delete", "Employee", new { id = e.Id })">De-
lete</a>
        </td>
    </tr>
}
</table>

```

**OUTPUT**



## Add Employee

Name:   
Department:   
Salary:

Save